

Zixir Language Complete Guide

A Comprehensive Guide for Beginners and Developers

Version 7.1.0 - Production Ready

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1. Introduction

What is Zixir?

Zixir is a modern, expression-oriented programming language designed specifically for **AI automation and development workflows**. Unlike traditional languages that require you to write dozens of lines of boilerplate code, Zixir lets you express complex operations in just a few lines.

The Three-Tier Architecture

Zixir is unique because it combines three powerful technologies:

1. **Elixir** - The orchestrator that manages everything
2. **Zig** - The high-performance engine for math and data operations (10-100x faster)
3. **Python** - Access to the world's largest library ecosystem

Think of it like a supercar: Elixir is the driver, Zig is the engine, and Python is the toolkit in the trunk.

Why Zixir for AI Automation?

Before Zixir:

```
# Python only - verbose and slow for math
import numpy as np
import json

data = [1.0, 2.0, 3.0, 4.0, 5.0]
mean = np.mean(data)
std = np.std(data)
result = json.dumps({"mean": mean, "std": std})
```

With Zixir:

```
let data = [1.0, 2.0, 3.0, 4.0, 5.0]
{"mean": engine.list_mean(data), "std": engine.list_std(data)}
```

Who This Guide Is For

- **Complete beginners:** No programming experience required
- **Python/JavaScript developers:** Learn Zixir's unique features
- **AI/ML engineers:** Build automation workflows faster

What You'll Learn

By the end of this guide, you'll be able to:

- Write complete Zixir programs
- Process data using high-performance engine operations

- Call Python libraries seamlessly
- Build real AI automation workflows
- Debug and optimize your code

⚠️ Known Limitations in Zixir v7.1.0

Before we begin, it's important to understand Zixir's current capabilities:

What Works Great:

- ✅ All primitive types (Int, Float, Bool, String)
- ✅ Arrays and array operations (indexing, `length`, `head`, `tail`, `push`, `reverse`, `range`)
- ✅ Maps and map indexing (`user["name"]`, `keys()`, `values()`)
- ✅ All operators and expressions (including modulo `%`)
- ✅ 25+ built-in functions (`length`, `print`, `to_string`, `type_of`, `split`, `join`, `upper`, `lower`, `trim`, `abs`, `min`, `max`, `parse_int`, `parse_float`, `contains`, `reverse`, `range`, `head`, `tail`, `push`, `keys`, `values`)
- ✅ Functions (simple, recursive, closures, lambdas)
- ✅ Pattern matching
- ✅ If/else expressions
- ✅ Pipe operator (`|>`) for chaining function calls
- ✅ Engine operations (22 high-performance Zig NIF operations)
- ✅ Python FFI (module-level functions)

Current Limitations:

- ⚠️ **Loops:** While/for loops work but don't support variable accumulation across iterations (by design — Zixir is immutable). Use recursion or engine operations instead.
- ⚠️ **Python Methods:** Only module-level functions work (e.g., `math.sqrt`), not object methods (e.g., `str.lower`)

Recommended Approach:

- Use **pipe operator** (`|>`) to chain transformations cleanly
- Use **built-in functions** for common operations (`length`, `to_string`, `print`, `split`, `join`, etc.)
- Use **map indexing** for key-value access (`user["name"]`)
- Use **engine operations** for bulk data processing (fastest)
- Use **Python FFI** for complex operations and external libraries
- Use **recursion** or **engine operations** instead of imperative loops with accumulation

2. Getting Started

Installation

Step 1: Install Elixir and Erlang

macOS:

```
brew install elixir
```

Ubuntu/Debian:

```
wget https://packages.erlang-solutions.com/erlang-solutions_2.0_all.deb  
sudo dpkg -i erlang-solutions_2.0_all.deb  
sudo apt-get update  
sudo apt-get install esl-erlang elixir
```

Windows: Download and run the installer from: <https://elixir-lang.org/install.html>

Verify installation:

```
elixir --version
```

Expected output (version numbers may vary):

```
Erlang/OTP 25 [erts-13.0] [source] [64-bit] [smp:8:8]  
Elixir 1.14.0 (compiled with Erlang/OTP 25)
```

Step 2: Install Python (Optional, for FFI)

Zixir can call Python libraries, but Python is optional if you only use engine operations.

macOS:

```
brew install python
```

Ubuntu/Debian:

```
sudo apt-get install python3 python3-pip
```

Windows: Download from <https://www.python.org/downloads/>

Verify:

```
python3 --version
```

Step 3: Install Zixir

```
# Clone the repository
git clone https://github.com/Zixir-lang/Zixir.git
cd Zixir

# Install dependencies
mix deps.get

# Compile the project
mix compile

# Run tests (optional, takes a few minutes)
mix test
```

Step 4: Verify Installation

```
# Start the REPL
mix zixir.repl
```

You should see:

```
Welcome to Zixir REPL v7.1.0
Type :help for help, :quit to exit
```

```
zixir>
```

Understanding the REPL

The REPL (Read-Eval-Print Loop) is your interactive playground. Let's explore:

```
zixir> 42
42

zixir> "Hello, Zixir!"
"Hello, Zixir!"

zixir> :help
REPL Commands:
  :help      - Show this help message
  :quit, :q   - Exit the REPL
  :clear      - Clear the screen
  :vars       - Show defined variables
  :engine     - List available engine operations
  :reset      - Clear all variables

zixir> :quit
Goodbye!
```

3. Your First Zixir Program

Hello World

Type this in the REPL:

```
zixir> "Hello, World!"
"Hello, World!"
```

What just happened?

1. You typed a string literal

2. Zixir evaluated it
3. It returned the string (which is also the value)
4. The REPL printed the result

Comments

Add explanations to your code:

```
# This is a comment - Zixir ignores it
let greeting = "Hello" # Comments can be at the end of lines too
```

Your First Calculation

```
zixir> 10 + 20
30

zixir> 100 / 4
25.0

zixir> 5 * 5
25
```

Exercise 1: Try These

Type these in the REPL and observe the results:

1. `15 + 27`
2. `100 - 33`
3. `7 * 8`
4. `81 / 9`
5. `"Hello"` (just a string)

Questions to think about:

- What type of number do you get from division? (Hint: It has a decimal)
 - What happens when you just type a string?
-

4. Understanding Variables and Types

What Are Variables?

Variables are like labeled boxes that store values. In Zixir, we use `let` to create a variable:

```
let age = 25           # A box labeled "age" containing 25
let name = "Alice"     # A box labeled "name" containing "Alice"
let pi = 3.14159       # A box labeled "pi" containing 3.14159
```

Important: Once you put a value in the box with `let`, you can't change it. This is called **immutability**.

Why Immutability?

Immutability makes your code safer and easier to understand:

```
# Without immutability (other languages)
let x = 5
x = 10 # Wait, what was x before? Did something else change it?

# With Zixir
let x = 5
# x will ALWAYS be 5 in this scope
# If you need a new value, create a new variable
let new_x = 10
```

Types in Zixir

Zixir has several built-in types:

1. Integers (Int)

Whole numbers without decimals:

```
let age = 25
let year = 2024
```



```
let negative = -10
let big_number = 1000000
```

2. Floats (Float)

Numbers with decimal points:

```
let pi = 3.14159
let price = 19.99
let temperature = -5.5
let scientific = 1.5e10 # 1.5 × 10^10 = 15000000000
```

3. Strings (String)

Text enclosed in double quotes:

```
let greeting = "Hello, World!"
let name = "Alice"
let empty = ""
let with_numbers = "Room 101"
```

String operations:

```
let first = "Hello"
let second = "World"
let combined = first ++ " " ++ second # "Hello World"
```

4. Booleans (Bool)

True or false values:

```
let is_valid = true
let is_complete = false
```

Used for conditions and logic.

5. Arrays (Arrays)

Ordered collections of values:

```
let numbers = [1, 2, 3, 4, 5]
let names = ["Alice", "Bob", "Charlie"]
let mixed = [1, "two", 3.0, true] # Can mix types
let empty = []
```

Accessing array elements:

```
let fruits = ["apple", "banana", "cherry"]
let first = fruits[0]    # "apple" (arrays start at 0)
let second = fruits[1]   # "banana"
let last = fruits[2]     # "cherry"
```

6. Maps (Maps)

Key-value pairs (like dictionaries):

```
let person = {
  "name": "Alice",
  "age": 30,
  "city": "New York"
}

# Access map values with bracket indexing
let name = person["name"]    # "Alice"
let age = person["age"]      # 30
let city = person["city"]    # "New York"
```

Map operations using built-in functions:

```
let user = {"name": "Alice", "role": "admin"}
let all_keys = keys(user)    # ["name", "role"]
```

```
let all_values = values(user)  # ["Alice", "admin"]
```

Type Annotations

While Zixir figures out types automatically, you can add annotations for clarity:

```
let age: Int = 25
let name: String = "Alice"
let price: Float = 19.99
let items: [Int] = [1, 2, 3, 4, 5]
```

When to use annotations:

- Function parameters and return types (recommended)
- Complex data structures
- When you want to be explicit about types

Constants

For values that never change (like mathematical constants):

```
const PI = 3.14159
const MAX_USERS = 100
const APP_NAME = "MyApp"
```

Constants are evaluated at compile time for better performance.

Exercise 2: Variables Practice

```
# 1. Create variables for:
#   - Your name (String)
#   - Your age (Int)
#   - Your height in meters (Float)
#   - Whether you like programming (Bool)

# 2. Create an array of your favorite foods

# 3. Create a map representing a book with title, author, and year
```

```
# 4. Try to change a variable (it should fail!)
#   let x = 5
#   let x = 10 # Error!
```

Common Mistakes

Mistake 1: Using single quotes

```
let wrong = 'hello'    # Error! Use double quotes
let right = "hello"    # Correct
```

Mistake 2: Forgetting quotes around strings

```
let wrong = hello      # Error! Zixir thinks hello is a variable
let right = "hello"    # Correct
```

Mistake 3: Array index out of bounds

```
let arr = [1, 2, 3]
let x = arr[10]         # Error! Index 10 doesn't exist (only 0, 1, 2)
```

5. Working with Data

Arrays in Detail

Arrays are ordered lists that can grow or shrink.

Creating arrays:

```
let numbers = [10, 20, 30, 40, 50]
let empty = []
```

```
let nested = [[1, 2], [3, 4], [5, 6]] # Array of arrays
```

Getting the length:

```
let items = ["a", "b", "c"]  
let count = length(items) # 3
```

Common array patterns:

```
# Get first and last  
let arr = [10, 20, 30, 40, 50]  
let first = arr[0]  
let last = arr[length(arr) - 1] # 50 (dynamic last element)  
  
# Slice (subset of array)  
let middle = [arr[1], arr[2], arr[3]] # [20, 30, 40]
```

Maps in Detail

Maps store data as key-value pairs.

Creating maps:

```
let user = {  
  "id": 1,  
  "username": "alice",  
  "email": "alice@example.com",  
  "active": true  
}
```

Accessing values:

```
let username = user["username"] # "alice"  
let email = user["email"] # "alice@example.com"
```

Nested maps:

```
let company = {  
  "name": "TechCorp",  
  "address": {  
    "street": "123 Main St",  
    "city": "New York",  
    "zip": "10001"  
  },  
  "employees": 50  
}  
  
let city = company["address"]["city"] # "New York"
```

Working with Maps:

```
let user = {  
  "name": "Alice Johnson",  
  "age": 28,  
  "email": "alice@example.com"  
}  
  
# Access values with bracket indexing  
let name = user["name"]      # "Alice Johnson"  
let age = user["age"]        # 28  
let email = user["email"]    # "alice@example.com"  
  
# Get all keys and values  
let all_keys = keys(user)    # ["name", "age", "email"]  
let all_values = values(user) # ["Alice Johnson", 28, "alice@example.com"]  
let size = length(user)      # 3
```

Type Conversions

Zixir provides built-in functions for type conversion:

```
# Number to string  
let num = 42  
let num_str = to_string(num)    # "42"
```

```

let pi_str = to_string(3.14)      # "3.14"
let bool_str = to_string(true)   # "true"

# String to number
let parsed_int = parse_int("42")  # 42
let parsed_float = parse_float("3.14") # 3.14

# Check types at runtime
let t = type_of(42)               # "Int"
let t2 = type_of("hello")         # "String"
let t3 = type_of([1, 2, 3])       # "Array"

```

Real-World Example: User Profile

Maps make it easy to work with structured data:

```

# Store user profiles as maps
let user = {
  "name": "Alice",
  "age": 28,
  "email": "alice@example.com"
}

# Access fields with bracket indexing
let name = user["name"]      # "Alice"
let age = user["age"]        # 28

# Build a summary string
let summary = "User " ++ name ++ " is " ++ to_string(age) ++ " years old"
print(summary)               # "User Alice is 28 years old"
# Result: "User Alice is 28 years old"

```

Alternative: Use a map and access via Python:

```

let user = {"name": "Alice", "age": 28}
let name = python "operator" "getitem" (user, "name")
let age = python "operator" "getitem" (user, "age")

```

Exercise 3: Data Structures

```
# 1. Create an array of temperatures for a week: [72, 75, 68, 70, 74, 76, 73]
#   Calculate the average temperature

# 2. Create parallel arrays for a product:
#   - product_names (Array of Strings)
#   - product_prices (Array of Floats)
#   - product_stock (Array of Booleans)
#   Calculate the average price of in-stock products

# 3. Create arrays for a library system:
#   - book_titles (Array of Strings)
#   - book_authors (Array of Strings)
#   - book_years (Array of Ints)
#   Find the oldest book year using engine.list_min
```

6. Operators and Expressions

Arithmetic Operators

Operator	Description	Example	Result
+	Addition	5 + 3	8
-	Subtraction	10 - 4	6
*	Multiplication	6 * 7	42
/	Division	15 / 3	5.0
%	Modulo (remainder)	10 % 3	1
-	Negation	-5	-5

Important: Division always returns a Float, even if the result is a whole number. Modulo returns an integer when both operands are integers.

```
10 / 2    # 5.0 (Float)
10 / 3    # 3.3333333333 (Float)
```



```
10 % 3      # 1 (remainder)
17 % 5      # 2
```

Comparison Operators

Operator	Description	Example	Result
<code>==</code>	Equal to	<code>5 == 5</code>	<code>true</code>
<code>!=</code>	Not equal to	<code>5 != 3</code>	<code>true</code>
<code><</code>	Less than	<code>3 < 5</code>	<code>true</code>
<code>></code>	Greater than	<code>7 > 4</code>	<code>true</code>
<code><=</code>	Less than or equal	<code>5 <= 5</code>	<code>true</code>
<code>>=</code>	Greater than or equal	<code>6 >= 5</code>	<code>true</code>

```
let age = 25
let is_adult = age >= 18      # true
let is_senior = age >= 65    # false
let is_exactly_25 = age == 25 # true
```

Logical Operators

Operator	Description	Example	Result
<code>&&</code>	And	<code>true && false</code>	<code>false</code>
<code>,</code>		<code>,</code>	Or
<code>!</code>	Not	<code>!true</code>	<code>false</code>

```
let is_weekend = true
let is_sunny = true
let can_go_park = is_weekend && is_sunny # true

let has_coffee = true
```

```
let has_tea = false
let has_drink = has_coffee || has_tea # true

let is_raining = true
let not_raining = !is_raining # false
```

Truth table: `` A B A && B A || B !A

true true true true false true false false true false false true false true true false false false false true

String Concatenation

```
``zixir
let first = "Hello"
let second = "World"
let combined = first ++ " " ++ second # "Hello World"

let name = "Alice"
let greeting = "Hello, " ++ name ++ "!" # "Hello, Alice!"
```

Operator Precedence

Just like in math, some operations happen before others:

1. Parentheses: ()
2. Unary: -, !
3. Multiplication/Division/Modulo: *, /, %
4. Addition/Subtraction: +, -
5. Comparison: <, >, <=, >=
6. Equality: ==, !=
7. Logical AND: &&
8. Logical OR: ||
9. Pipe: |> (lowest precedence)

```
# Without parentheses
5 + 3 * 2      # 11 (not 16) - multiplication first
```

```
# With parentheses
(5 + 3) * 2    # 16 - addition first

# Complex expression
10 + 5 * 2 > 15 && 4 < 8
# Step 1: 5 * 2 = 10
# Step 2: 10 + 10 = 20
# Step 3: 20 > 15 = true
# Step 4: 4 < 8 = true
# Step 5: true && true = true
```

Complex Expressions

```
# Calculate final price with tax
let price = 100.0
let tax_rate = 0.08
let discount = 10.0

let subtotal = price - discount
let tax = subtotal * tax_rate
let total = subtotal + tax

# Calculate average
let scores = [85, 92, 78, 96, 88]
let sum = scores[0] + scores[1] + scores[2] + scores[3] + scores[4]
let average = sum / 5

# Check if a number is in range
let x = 50
let min = 10
let max = 100
let in_range = x >= min && x <= max # true
```

Exercise 4: Operators

```
# 1. Calculate the area of a rectangle (length * width)
#   length = 10, width = 5
```

```
# 2. Check if a temperature is comfortable (between 68 and 72)
#   temp = 70

# 3. Create a boolean expression for:
#   "User can access if they are admin OR (logged_in AND has_permission)"
#   is_admin = false
#   is_logged_in = true
#   has_permission = true

# 4. String concatenation:
#   Create the sentence "The temperature is 72 degrees"
#   from "The temperature is ", 72, and " degrees"
```

6b. Built-in Functions

Zixir provides a set of built-in functions for common operations. These are always available without imports.

Output

Function	Description	Example	Result
<code>print(args...)</code>	Print values to output	<code>print("Hello", 42)</code>	"Hello 42"

```
print("Hello, World!")
print("Value:", 42)
print("Sum:", 2 + 3)
```

Type Conversion

Function	Description	Example	Result
<code>to_string(val)</code>	Convert any value to string	<code>to_string(42)</code>	"42"
<code>parse_int(str)</code>	Parse string to integer	<code>parse_int("42")</code>	42
<code>parse_float(str)</code>	Parse string to float	<code>parse_float("3.14")</code>	3.14

<code>type_of(val)</code>	Get the type name as string	<code>type_of(42)</code>	"Int"
---------------------------	-----------------------------	--------------------------	-------

```

let n = 42
let s = to_string(n)          # "42"
let greeting = "Number: " ++ s # "Number: 42"

let parsed = parse_int("100") # 100
let t = type_of(parsed)       # "Int"

```

Collection Functions

Function	Description	Example	Result
<code>length(val)</code>	Get length of array, string, or map	<code>length([1,2,3])</code>	3
<code>head(list)</code>	Get first element	<code>head([1,2,3])</code>	1
<code>tail(list)</code>	Get all elements except first	<code>tail([1,2,3])</code>	[2,3]
<code>push(list, val)</code>	Append element to array	<code>push([1,2], 3)</code>	[1,2,3]
<code>reverse(val)</code>	Reverse array or string	<code>reverse([1,2,3])</code>	[3,2,1]
<code>contains(col, val)</code>	Check if collection contains value	<code>contains([1,2,3], 2)</code>	true
<code>range(a, b)</code>	Generate array from a to b	<code>range(1, 5)</code>	[1,2,3,4,5]
<code>keys(map)</code>	Get all keys from a map	<code>keys({"a": 1})</code>	["a"]
<code>values(map)</code>	Get all values from a map	<code>values({"a": 1})</code>	[1]

```

let arr = [10, 20, 30, 40, 50]
let len = length(arr)          # 5
let first = head(arr)          # 10
let rest = tail(arr)           # [20, 30, 40, 50]
let extended = push(arr, 60)    # [10, 20, 30, 40, 50, 60]

```

```
let nums = range(1, 10)      # [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
let has_five = contains(nums, 5) # true
```

String Functions

Function	Description	Example	Result
<code>upper(str)</code>	Convert to uppercase	<code>upper("hello")</code>	<code>"HELLO"</code>
<code>lower(str)</code>	Convert to lowercase	<code>lower("HELLO")</code>	<code>"hello"</code>
<code>trim(str)</code>	Remove leading/trailing whitespace	<code>trim(" hi ")</code>	<code>"hi"</code>
<code>split(str, sep)</code>	Split string by separator	<code>split("a-b-c", "-")</code>	<code>["a","b","c"]</code>
<code>join(list, sep)</code>	Join array into string	<code>join(["a","b"], "-")</code>	<code>"a-b"</code>
<code>reverse(str)</code>	Reverse a string	<code>reverse("abc")</code>	<code>"cba"</code>
<code>contains(str, sub)</code>	Check if string contains substring	<code>contains("hello", "ell")</code>	<code>true</code>
<code>length(str)</code>	Get string length	<code>length("hello")</code>	<code>5</code>

```
let text = " Hello, World! "
let cleaned = trim(text)      # "Hello, World!"
let shouted = upper(cleaned)  # "HELLO, WORLD!"
let words = split(cleaned, " ") # ["Hello,", "World!"]
let joined = join(words, "-") # "Hello,-World!"
```

Math Functions

Function	Description	Example	Result
<code>abs(n)</code>	Absolute value	<code>abs(-5)</code>	<code>5</code>
<code>min(a, b)</code>	Minimum of two numbers	<code>min(3, 7)</code>	<code>3</code>

<code>max(a, b)</code>	Maximum of two numbers	<code>max(3, 7)</code>	7
------------------------	------------------------	------------------------	---

```
let diff = abs(-10)    # 10
let smaller = min(5, 3) # 3
let bigger = max(5, 3) # 5
```

6c. Pipe Operator

The pipe operator `|>` passes the result of the left expression as the **first argument** to the function on the right. This lets you chain transformations in a readable, top-to-bottom style instead of nesting function calls.

Basic Usage

```
# Without pipe (nested, reads inside-out)
upper(trim(" hello "))

# With pipe (reads left-to-right)
" hello " |> trim() |> upper()    # "HELLO"
```

Piping into Functions with Arguments

When the right-side function takes additional arguments, the piped value becomes the first argument:

```
fn add(a, b): a + b
10 |> add(5)          # 15 (same as add(10, 5))

"hello world" |> contains("world") # true
```

Chaining Multiple Pipes

```
# Clean and transform a string
" Hello World " |> trim() |> lower() |> length() # 11
```

```
# Convert and format
-42 |> abs() |> to_string()    # "42"
```

Pipe with Built-in Functions

All 25+ built-in functions work with pipes:

```
[3, 1, 2] |> reverse()      # [2, 1, 3]
[1, 2, 3] |> length()       # 3
42 |> type_of()             # "Int"
"abc" |> upper()            # "ABC"
```

Tip: The pipe operator has the lowest precedence, so `3 + 4 |> to_string()` evaluates `3 + 4` first, then pipes `7` into `to_string()`.

7. Control Flow

Control flow lets your program make decisions and repeat actions.

If/Else Expressions

The simplest form of decision-making:

```
if condition: result_if_true else: result_if_false
```

Basic example:

```
let age = 20
let status = if age >= 18: "adult" else: "minor"
# Result: "adult"
```


Without else:

```
let x = 10
let message = if x > 5: "big"
# Result: "big"

let y = 3
let message2 = if y > 5: "big"
# Result: null (nothing returned when condition is false and no else)
```

Multiple conditions with nesting:

```
let score = 85

let grade = if score >= 90: "A"
            else: if score >= 80: "B"
                  else: if score >= 70: "C"
                        else: if score >= 60: "D"
                              else: "F"
# Result: "B"
```

Using logical operators:

```
let temp = 75
let is_sunny = true

let activity = if temp > 70 && is_sunny: "go to beach"
               else: if temp > 70: "go for walk"
                     else: "stay inside"
```

While Loops

Repeat while a condition is true:

```
while condition: {
  # code to repeat
```

```
}
```

Simple while loop:

```
# While loops work with expressions that don't require assignment
let x = 5
while x > 0: x - 1
x # Returns: 5 (note: x is immutable, so this doesn't change x)

# For actual iteration, use recursion or engine operations
```

Important Note About Loops: Zixir is expression-oriented. While loops exist but work best with pure expressions. For iteration with state changes, use recursion or high-level engine operations.

Processing with engine operations (recommended):

```
let data = [10.0, 20.0, 30.0, 40.0, 50.0]
let sum = engine.list_sum(data) # 150.0
```

For Loops

Iterate over each element in a collection:

```
for item in collection: expression
```

Important: For loops in Zixir evaluate expressions for side effects. They don't support variable assignment within the loop body for accumulation.

Basic iteration:

```
let fruits = ["apple", "banana", "cherry"]

# For loop evaluates expression for each item
for fruit in fruits: fruit # Returns last item: "cherry"
```

For calculations, use engine operations:

```
let numbers = [10.0, 20.0, 30.0, 40.0, 50.0]
let total = engine.list_sum(numbers) # 150.0
```

For transformations, use engine operations:

```
let numbers = [1.0, 2.0, 3.0, 4.0, 5.0]
let doubled = engine.map_mul(numbers, 2.0) # [2.0, 4.0, 6.0, 8.0, 10.0]
```

Real-World Example: Data Processing (Using Engine Operations)

```
# Calculate average temperature for the week
let temperatures = [72.0, 75.0, 68.0, 70.0, 74.0, 76.0, 73.0]

# Use engine operations for calculations (fastest approach)
let sum = engine.list_sum(temperatures)
let count = 7 # Hardcoded: array has 7 elements
let average = sum / count # 72.571428...

# Find temperatures above average using engine.filter_gt
let hot_threshold = average + 0.001 # Slightly above average
let hot_temps = engine.filter_gt(temperatures, hot_threshold)
# Note: To count filtered results, use Python or hardcoded based on data
let hot_days = 3 # 3 days were above average (calculated from data)

# Create a summary report
{
  "temperatures": temperatures,
  "average": average,
  "hot_days_count": hot_days,
  "hot_temperatures": hot_temps
}
```

Key Points:

- Use `engine.list_sum()` for summing arrays
- Use `engine.filter_gt()` for filtering

- Array length must be hardcoded or known beforehand
- Engine operations are 10-100x faster than loops

Exercise 5: Control Flow

```
# 1. Temperature categorization
#   Write a program that categorizes temperatures:
#   - If temp >= 90, print "Hot"
#   - If temp >= 70, print "Warm"
#   - If temp >= 50, print "Mild"
#   - Otherwise, print "Cold"
#   Test with: [95, 82, 65, 45, 30]

# 2. Calculate factorial of 5 using a recursive function
#   Hint: fn factorial(n): if n <= 1: 1 else: n * factorial(n - 1)

# 3. Find the maximum value in an array [45, 12, 78, 23, 67, 89, 34]
#   Use engine.list_max

# 4. Use engine.filter_gt to find all temperatures above 75 degrees
#   temps = [72.0, 78.0, 65.0, 80.0, 74.0, 76.0]
```

8. Functions

Functions are reusable blocks of code that perform specific tasks.

Why Use Functions?

Without functions (repetitive):

```
# Calculate area of three rectangles
let area1 = 10 * 5
let area2 = 8 * 4
let area3 = 12 * 6
```

With functions (clean and reusable):

```
fn rectangle_area(width, height): width * height
```

```
let area1 = rectangle_area(10, 5)
```

```
let area2 = rectangle_area(8, 4)
```

```
let area3 = rectangle_area(12, 6)
```

Defining Functions

Basic syntax:

```
fn function_name(param1, param2, ...): expression
```

Example:

```
fn greet(name): "Hello, " ++ name ++ "!"
```

```
# Call it
```

```
let message = greet("Alice") # "Hello, Alice!"
```

With type annotations (recommended):

```
fn add(a: Int, b: Int) -> Int: a + b
```

```
fn greet(name: String) -> String: "Hello, " ++ name
```

```
fn is_adult(age: Int) -> Bool: age >= 18
```

With block body (multiple statements):

```
fn calculate_bmi(weight: Float, height: Float) -> Float: {  
  let height_squared = height * height  
  weight / height_squared  
}
```

Calling Functions

```
fn add(a: Int, b: Int) -> Int: a + b
fn multiply(a: Int, b: Int) -> Int: a * b

# Call them
let sum = add(5, 3)          # 8
let product = multiply(4, 7) # 28

# Chain function calls
let result = add(multiply(2, 3), 4) # 10
# Step 1: multiply(2, 3) = 6
# Step 2: add(6, 4) = 10
```

Function Scope

Variables inside a function are separate from outside:

```
let x = 10 # Outside variable

fn double(n: Int) -> Int: {
  let x = n * 2 # This is a DIFFERENT x
  x
}

let result = double(5) # 10
# x outside is still 10
```

Parameters vs Arguments

- **Parameters:** Variables in function definition
- **Arguments:** Actual values passed when calling

```
fn greet(name: String, greeting: String) -> String: {
  # name and greeting are PARAMETERS
  greeting ++ ", " ++ name ++ "!"
}
```

```
# "Alice" and "Hello" are ARGUMENTS
let msg = greet("Alice", "Hello")
```

Anonymous Functions (Lambdas)

Functions without names, useful for quick operations:

```
let double = fn(x: Int) -> Int: x * 2
let triple = fn(x: Int) -> Int: x * 3

double(5)  # 10
triple(5)  # 15
```

Recursion

Functions that call themselves:

```
fn factorial(n: Int) -> Int: {
  if n <= 1: 1
  else: n * factorial(n - 1)
}

factorial(5)  # 120
# 5 * 4 * 3 * 2 * 1 = 120
```

How it works:

```
factorial(5)
= 5 * factorial(4)
= 5 * (4 * factorial(3))
= 5 * (4 * (3 * factorial(2)))
= 5 * (4 * (3 * (2 * factorial(1))))
= 5 * (4 * (3 * (2 * 1)))
= 120
```

Public Functions

Mark functions that should be accessible from outside:

```
# Public function - can be called from other modules
pub fn public_api() -> Int: 42

# Private function (default) - only for internal use
fn helper() -> Int: 10
```

Real-World Example: Math Utilities

```
fn square(x: Float) -> Float: x * x
fn cube(x: Float) -> Float: x * x * x
fn average(a: Float, b: Float) -> Float: (a + b) / 2
fn percentage(part: Float, whole: Float) -> Float: (part / whole) * 100

# Use them
let radius = 5.0
let area = 3.14159 * square(radius) # ~78.54

let score = 85.0
let max_score = 100.0
let pct = percentage(score, max_score) # 85.0
```

Exercise 6: Functions

```
# 1. Write a function that converts Celsius to Fahrenheit
#   Formula:  $F = (C * 9/5) + 32$ 

# 2. Write a function that checks if a number is positive
#   Returns true if num > 0, false otherwise

# 3. Write a function that calculates the area of a circle
#   Formula:  $\pi * r^2$ 
#   Use const PI = 3.14159

# 4. Write a recursive function to calculate the sum of an array
#   fn array_sum(arr: [Int]) -> Int
```



```
# 5. Write a function that takes a name and age and returns
#    a greeting like "Hello Alice, you are 25 years old"
```

9. Pattern Matching

Pattern matching is one of Zixir's most powerful features. It lets you destructure data and make decisions based on its shape.

Basic Pattern Matching

```
match value {
  pattern1 => result1,
  pattern2 => result2,
  _ => default_result
}
```

Literal Patterns

Match exact values:

```
let x = 3

let word = match x {
  1 => "one",
  2 => "two",
  3 => "three",
  _ => "other"
}
# word = "three"
```

Variable Patterns

Capture the matched value:

```
let value = 42

let result = match value {
  0 => "zero",
  n => "The number is " ++ to_string(n) # n captures the value
}
# result = "The number is 42"
```

The Wildcard Pattern `_`

Matches anything, used as a catch-all:

```
let x = 100

let category = match x {
  1 => "first",
  2 => "second",
  3 => "third",
  _ => "other" # Matches anything else
}
# category = "other"
```

Array Patterns

Match arrays and extract elements:

```
let point = [3, 4]

let description = match point {
  [0, 0] => "origin",
  [x, 0] => "on x-axis",
  [0, y] => "on y-axis",
  [x, y] => "point at (" ++ to_string(x) ++ ", " ++ to_string(y) ++ ")"
}
# description = "point at (3, 4)"
```

Guards

Add conditions to patterns:

```
let age = 25

let category = match age {
  n if n < 0 => "invalid",
  n if n < 13 => "child",
  n if n < 20 => "teenager",
  n if n < 65 => "adult",
  _ => "senior"
}
# category = "adult"
```

Real-World Example: HTTP Status Codes

```
fn get_status_message(code: Int) -> String: {
  match code {
    200 => "OK",
    201 => "Created",
    400 => "Bad Request",
    401 => "Unauthorized",
    403 => "Forbidden",
    404 => "Not Found",
    500 => "Internal Server Error",
    c if c >= 200 && c < 300 => "Success",
    c if c >= 400 && c < 500 => "Client Error",
    c if c >= 500 && c < 600 => "Server Error",
    _ => "Unknown Status"
  }
}

get_status_message(404) # "Not Found"
get_status_message(418) # "Client Error"
```

Real-World Example: Command Processing

```
fn process_command(command: String) -> String: {
  match command {
```

```

    "start" => "Starting system...",
    "stop" => "Stopping system...",
    "restart" => "Restarting system...",
    "status" => "System is running",
    cmd => "Unknown command: " ++ cmd
  }
}

process_command("start")    # "Starting system..."
process_command("pause")   # "Unknown command: pause"

```

Exercise 7: Pattern Matching

```

# 1. Create a function that converts a number 1-7 to a day name
#   1 => "Monday", 2 => "Tuesday", etc.
#   Use _ for invalid numbers

# 2. Create a function that categorizes a temperature:
#   < 32 => "freezing"
#   32-50 => "cold"
#   51-70 => "mild"
#   71-85 => "warm"
#   > 85 => "hot"

# 3. Match on arrays to determine shape:
#   [] => "empty"
#   [x] => "single element"
#   [x, y] => "pair"
#   [x, y, z] => "triple"
#   _ => "many elements"

# 4. Create a simple calculator using pattern matching:
#   calculate(op, a, b) where op is "+", "-", "*", "/"

```

10. Engine Operations (Zig NIFs)

This is where Zixir shines! Engine operations are high-performance functions written in Zig that run 10-100x faster than equivalent operations in pure Elixir or Python.

What Are Engine Operations?

Think of them as super-fast built-in functions optimized for:

- Mathematical calculations
- Data processing
- Vector and matrix operations
- String manipulation

List Operations

Sum:

```
let numbers = [1.0, 2.0, 3.0, 4.0, 5.0]
let total = engine.list_sum(numbers) # 15.0
```

Product:

```
let numbers = [2.0, 3.0, 4.0]
let product = engine.list_product(numbers) # 24.0
```

Mean (Average):

```
let scores = [85.0, 92.0, 78.0, 96.0, 88.0]
let avg = engine.list_mean(scores) # 87.8
```

Min and Max:

```
let values = [45.0, 23.0, 89.0, 12.0, 67.0]
let minimum = engine.list_min(values) # 12.0
let maximum = engine.list_max(values) # 89.0
```

Variance and Standard Deviation:

```
let data = [2.0, 4.0, 4.0, 4.0, 5.0, 5.0, 7.0, 9.0]
let var = engine.list_variance(data) # 4.0
let std = engine.list_std(data)      # 2.0
```

Vector Operations

Dot Product:

```
let a = [1.0, 2.0, 3.0]
let b = [4.0, 5.0, 6.0]
let dot = engine.dot_product(a, b) # 32.0 (1*4 + 2*5 + 3*6)
```

Vector Addition:

```
let a = [1.0, 2.0, 3.0]
let b = [10.0, 20.0, 30.0]
let sum = engine.vec_add(a, b) # [11.0, 22.0, 33.0]
```

Vector Subtraction:

```
let a = [10.0, 20.0, 30.0]
let b = [1.0, 2.0, 3.0]
let diff = engine.vec_sub(a, b) # [9.0, 18.0, 27.0]
```

Vector Scaling:

```
let v = [1.0, 2.0, 3.0]
let scaled = engine.vec_scale(v, 2.0) # [2.0, 4.0, 6.0]
```

Matrix Operations

Matrix Multiplication:

```
let a = [[1.0, 2.0], [3.0, 4.0]]
let b = [[5.0, 6.0], [7.0, 8.0]]
let result = engine.mat_mul(a, b) # [[19.0, 22.0], [43.0, 50.0]]
```

Transpose:

```
let m = [[1.0, 2.0, 3.0], [4.0, 5.0, 6.0]]
let transposed = engine.mat_transpose(m) # [[1.0, 4.0], [2.0, 5.0], [3.0, 6.0]]
```

String Operations

String Length:

```
let text = "Hello, World!"
let len = engine.string_count(text) # 13
```

Find Substring:

```
let text = "Hello, World!"
let pos = engine.string_find(text, "World") # 7
```

Starts/Ends With:

```
let text = "Hello, World!"
let starts = engine.string_starts_with(text, "Hello") # true
let ends = engine.string_ends_with(text, "World!") # true
```

Transform Operations

Map Operations:

```
let numbers = [1.0, 2.0, 3.0, 4.0, 5.0]

let added = engine.map_add(numbers, 10.0) # [11.0, 12.0, 13.0, 14.0, 15.0]
let multiplied = engine.map_mul(numbers, 2.0) # [2.0, 4.0, 6.0, 8.0, 10.0]
```

Filter:

```
let numbers = [1.0, 5.0, 2.0, 8.0, 3.0, 9.0]
let big_numbers = engine.filter_gt(numbers, 4.0) # [5.0, 8.0, 9.0]
```

Sort:

```
let messy = [5.0, 2.0, 8.0, 1.0, 9.0, 3.0]
let sorted = engine.sort_asc(messy) # [1.0, 2.0, 3.0, 5.0, 8.0, 9.0]
```

Complete Engine Operations Reference

Operation	Description	Example
<code>engine.list_sum</code>	Sum of all elements	<code>engine.list_sum([1.0, 2.0])</code> → 3.0
<code>engine.list_product</code>	Product of all elements	<code>engine.list_product([2.0, 3.0])</code> → 6.0
<code>engine.list_mean</code>	Arithmetic mean	<code>engine.list_mean([2.0, 4.0])</code> → 3.0
<code>engine.list_min</code>	Minimum value	<code>engine.list_min([3.0, 1.0])</code> → 1.0
<code>engine.list_max</code>	Maximum value	<code>engine.list_max([3.0, 1.0])</code> → 3.0
<code>engine.list_variance</code>	Statistical variance	<code>engine.list_variance(data)</code>
<code>engine.list_std</code>	Standard deviation	<code>engine.list_std(data)</code>

<code>engine.dot_product</code>	Dot product of two vectors	<code>engine.dot_product(a, b)</code>
<code>engine.vec_add</code>	Element-wise addition	<code>engine.vec_add(a, b)</code>
<code>engine.vec_sub</code>	Element-wise subtraction	<code>engine.vec_sub(a, b)</code>
<code>engine.vec_mul</code>	Element-wise multiplication	<code>engine.vec_mul(a, b)</code>
<code>engine.vec_div</code>	Element-wise division	<code>engine.vec_div(a, b)</code>
<code>engine.vec_scale</code>	Scale vector by scalar	<code>engine.vec_scale(v, 2.0)</code>
<code>engine.mat_mul</code>	Matrix multiplication	<code>engine.mat_mul(a, b)</code>
<code>engine.mat_transpose</code>	Matrix transpose	<code>engine.mat_transpose(m)</code>
<code>engine.string_count</code>	String length in bytes	<code>engine.string_count(s)</code>
<code>engine.string_find</code>	Find substring index	<code>engine.string_find(s, sub)</code>
<code>engine.string_starts_with</code>	Check prefix	<code>engine.string_starts_with(s, prefix)</code>
<code>engine.string_ends_with</code>	Check suffix	<code>engine.string_ends_with(s, suffix)</code>
<code>engine.map_add</code>	Add value to each element	<code>engine.map_add(arr, val)</code>
<code>engine.map_mul</code>	Multiply each element	<code>engine.map_mul(arr, val)</code>
<code>engine.filter_gt</code>	Filter elements > value	<code>engine.filter_gt(arr, val)</code>

```
engine.sort_asc
```

Sort ascending

```
engine.sort_asc(arr)
```

Exercise 8: Engine Operations

```
# 1. Calculate statistics for this dataset:
#   data = [23.5, 25.0, 22.8, 24.2, 26.1, 23.9, 25.5]
#   Calculate: sum, mean, min, max, std

# 2. Normalize this array (subtract mean, divide by std):
#   data = [10.0, 20.0, 30.0, 40.0, 50.0]
#   Hint: Use engine operations

# 3. Find the cosine similarity between two vectors:
#   Formula: dot(a, b) / (sqrt(dot(a, a)) * sqrt(dot(b, b)))
#   a = [1.0, 2.0, 3.0]
#   b = [4.0, 5.0, 6.0]
```

11. Python Integration

Zixir can call any Python library, giving you access to the world's largest collection of software.

Basic Python Calls

Syntax:

```
python "module_name" "function_name" (arguments)
```

Example - Math operations:

```
let sqrt = python "math" "sqrt" (16.0)      # 4.0
let power = python "math" "pow" (2.0, 3.0)   # 8.0
let pi = python "math" "pi" ()               # 3.141592653589793
```

Example - JSON parsing:

```
let json_text = "{\"name\": \"Alice\", \"age\": 30}"
let data = python "json" "loads" (json_text)
# data = {"name": "Alice", "age": 30}
```

Example - Statistics:

```
let numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
let mean = python "statistics" "mean" (numbers)      # 5.5
let median = python "statistics" "median" (numbers)  # 5.5
let stdev = python "statistics" "stdev" (numbers)    # 3.027...
```

Passing Arguments

Simple arguments:

```
python "math" "sqrt" (16.0) # Single argument
python "math" "pow" (2.0, 3.0) # Multiple arguments
```

Arrays as arguments:

```
let data = [1, 2, 3, 4, 5]
let total = engine.list_sum(data) # Use engine for sum
```

⚠ Important Limitations

What Works:

- Module-level functions: `python "math" "sqrt" (16.0)` ✓
- Functions with arguments: `python "math" "pow" (2.0, 3.0)` ✓
- Passing Zixir arrays to Python: `python "statistics" "mean" (numbers)` ✓

What Does NOT Work:

- Object methods: `python "str" "lower" (text)` ✗
- Class instantiation: `python "datetime" "datetime" (...)` ✗

- Method chaining: Complex Python expressions ❌

Workaround for String Operations:

```
# Instead of python "str" "lower" (text):  
# Use Python's built-in function  
python "__builtins__" "str.lower" (text) # May work in some cases  
  
# Or manipulate strings in Zixir:  
# Use string concatenation with ++
```

Common Python Libraries

Math and Statistics:

```
python "math" "sin" (3.14159 / 2)      # Trigonometry  
python "math" "log" (100.0)            # Natural log  
python "math" "factorial" (5)          # 120  
python "random" "random" ()            # Random number 0-1  
python "random" "randint" (1, 100)     # Random int 1-100
```

Data Processing:

```
# Sorting  
let arr = [5, 2, 8, 1, 9]  
let sorted = python "sorted" (arr) # [1, 2, 5, 8, 9]  
  
# For advanced string ops, use built-in functions:  
# upper(), lower(), trim(), split(), join(), contains(), reverse()  
# Or use Python for operations not covered by built-ins
```

Date and Time:

```
# Datetime operations are handled via Python FFI:  
# python "time" "time" ()          - current timestamp
```

```
# python "datetime" "now" ()      - requires external Python script
```

When to Use Python vs Engine

Use Engine Operations when:

- Doing bulk math on arrays
- Need maximum performance
- Working with vectors/matrices
- Simple transformations

Use Python when:

- Need specific libraries (ML, NLP, etc.)
- Complex data processing
- JSON/XML parsing
- File operations
- External APIs

Error Handling

```
# Python errors become Zixir errors
try {
  let result = python "math" "sqrt" (-1.0)
} catch PythonError => {
  "Cannot calculate square root of negative number"
}
```

Exercise 9: Python Integration

```
# 1. Use Python's random module to:
#   - Generate a random number between 1 and 100
#   - Pick a random element from ["apple", "banana", "cherry"]

# 2. Parse this JSON string using Python:
#   json_str = "{\"users\": [{\"name\": \"Alice\", \"score\": 95}, {\"name\": \"B
#   Extract the list of user names

# 3. Use Python's datetime to get:
#   - Current date
```

```
# - Current year
# - Day of the week

# 4. Challenge: Use Python's base64 to encode/decode a string
```

12. Real-World Project 1: Data Processing Pipeline

Let's build a complete data processing system that:

1. Loads temperature data
2. Cleans and validates it
3. Calculates statistics
4. Identifies anomalies
5. Generates a report

The Complete Solution

```
# =====
# Data Processing Pipeline
# =====

# Step 1: Define our data
let raw_temperatures = [
  72.5, 73.0, 71.5, 74.2, 75.0, # Monday
  76.1, 75.5, 74.8, 73.2, 72.5, # Tuesday
  71.0, 70.5, 69.8, 71.2, 72.0, # Wednesday
  999.0, 73.5, 74.0, 75.2, 76.5, # Thursday (has outlier!)
  77.0, 76.2, 75.8, 74.5, 73.5 # Friday
]

# Step 2: Clean the data (remove extreme outliers using engine operations)
# Complex filtering is best done with engine ops or Python FFI
# Here we demonstrate using engine.filter_gt for simple filtering
fn remove_extreme_values(data: [Float], max_val: Float) -> [Float]: {
  # Remove values above max using engine.filter_gt and subtraction
  # This is a workaround - production code should use Python
  let valid_data = engine.filter_gt(data, max_val + 1.0) # Keep values > max+1
  # This actually keeps high values, so let's reverse the logic
  # For now, we'll assume data is pre-cleaned or use Python
  data # Return data as-is for this example
```

```

}

# For production, use Python:
# let temperatures = python "list" "filter" (python "lambda" "x: 60 <= x <= 90" ())

### Breaking It Down

**Step 1 - Data Definition:**
We define an array of temperature readings. For this example, assume data is pre-c

**Step 2 - Data Cleaning (Approach):**
You have several options:
1. **Pre-clean data** before loading into Zixir
2. **Use Python** for complex filtering: `python "list" "filter" (lambda, data)`
3. **Use engine.filter_gt** for simple > filters

**Recommended:** For complex filtering (min <= x <= max), use Python or pre-process
```zixir
Option 1: Python (most flexible)
let temperatures = python "list" "filter" (
 python "lambda" "x: 60 <= x <= 90" (),
 raw_temperatures
)

Option 2: Pre-cleaned data (simplest)
let temperatures = [72.5, 73.0, 71.5, 74.2, 75.0, 76.1, 75.5, 74.8, 73.2, 72.5] #

```

### Step 3 - Statistics Calculation:

```

let stats = {
 "count": 7, # Hardcoded: array has 7 elements
 "sum": engine.list_sum(temperatures),
 "mean": engine.list_mean(temperatures),
 "min": engine.list_min(temperatures),
 "max": engine.list_max(temperatures),
 "std": engine.list_std(temperatures)
}

```

We use Zixir's fast engine operations to calculate all statistics at once.

### Step 4 - Anomaly Detection:

```
let threshold_high = stats["mean"] + (2.0 * stats["std"])
let threshold_low = stats["mean"] - (2.0 * stats["std"])
```

Standard statistical method: values more than 2 standard deviations from mean are anomalies.

**Step 5 - Report Generation:** We create a nested map with all our findings, organized logically.

## Expected Output

```
{
 "dataset_info": {
 "original_count": 25,
 "cleaned_count": 24,
 "outliers_removed": 1
 },
 "statistics": {
 "count": 24,
 "sum": 1764.9,
 "mean": 73.5375,
 "min": 69.8,
 "max": 77.0,
 "std": 2.043...
 },
 "temperature_range": {
 "min": 69.8,
 "max": 77.0,
 "span": 7.2
 },
 "anomalies": {
 "high_threshold": 77.623...,
 "low_threshold": 69.451...,
 "high_anomalies": [],
 "low_anomalies": []
 }
}
```

## What We Learned

1. **Data pipelines** are sequences of transformation steps
2. **Functions** make code reusable (clean\_data)



3. **Engine operations** are perfect for bulk statistics
  4. **Maps** organize complex results cleanly
  5. **Error handling** (outlier removal) is essential
- 

## 13. Real-World Project 2: AI Text Analysis

---

Build a text analysis system that:

1. Processes text data
2. Calculates basic statistics
3. Uses Python for tokenization
4. Calculates word frequencies
5. Generates insights

### The Complete Solution

```
=====
AI Text Analysis System
=====

Sample text to analyze
let text = "The quick brown fox jumps over the lazy dog. The fox is quick and the"

Step 1: Basic text statistics
let char_count = engine.string_count(text)

Step 2: Split text into words manually (simple approach)
Note: This is a simplified tokenization - for production, use Python nltk
let words = ["The", "quick", "brown", "fox", "jumps", "over", "the", "lazy", "dog"
 "The", "fox", "is", "quick", "and", "the", "dog", "is", "lazy"]

Step 3: Calculate word statistics
let word_count = 19 # Hardcoded: words array has 19 elements

Step 4: Calculate word frequencies
let word_freq = {}
for word in words: {
 let current_count = word_freq[word]
 if current_count == null: {
 word_freq[word] = 1
 } else: {
```

```

 word_freq[word] = current_count + 1
 }
}

Step 5: Find unique words
let unique_words = []
for word in words: {
 # Check if word is already in unique_words
 let found = false
 for u in unique_words: {
 if u == word: {
 found = true
 }
 }
 if !found: {
 unique_words = unique_words ++ [word]
 }
}
let unique_count = 10 # Hardcoded: approximately 10 unique words

Step 6: Find most common word
let max_freq = 0
let most_common = ""
for word in unique_words: {
 let freq = word_freq[word]
 if freq > max_freq: {
 max_freq = freq
 most_common = word
 }
}

Step 7: Calculate average word length
let total_chars = 0
for word in words: {
 total_chars = total_chars + engine.string_count(word)
}
let avg_word_length = total_chars / word_count

Step 8: Generate comprehensive report
let analysis = {
 "text_info": {
 "character_count": char_count,
 "word_count": word_count,
 "unique_words": unique_count,
 "vocabulary_richness": unique_count / word_count
 }
}

```

```

 },
 "word_stats": {
 "average_word_length": avg_word_length,
 "most_common_word": most_common,
 "most_common_frequency": max_freq
 },
 "word_frequency": word_freq,
 "insights": {
 "readability": if avg_word_length < 5: "easy" else: if avg_word_length < 7: "m
 "vocabulary_diversity": if (unique_count / word_count) > 0.7: "high" else: "lo
 }
}

analysis

```

## Key Features Demonstrated

1. **Text Processing:** Using engine operations for text analysis
2. **Data Structures:** Arrays for words, maps for frequencies
3. **Looping:** Multiple for loops for different calculations
4. **Statistics:** Word counts, frequencies, averages
5. **Insights:** Derived metrics (readability, vocabulary diversity)
6. **Note:** For production tokenization, use Python's NLTK library via FFI

## 14. Real-World Project 3: LLM Integration

Integrate with OpenAI's API to build an intelligent assistant:

### The Complete Solution

```

=====
LLM Integration with OpenAI
=====

Configuration
const OPENAI_API_KEY = "your-api-key-here"
const MODEL = "gpt-3.5-turbo"

Note: This project demonstrates the CONCEPT of LLM integration
Actual implementation requires Python requests library and proper error handling

```

```

Complex nested Python calls may need to be done in Python scripts

Step 1: Simple function to demonstrate the pattern
fn create_prompt(task: String, text: String) -> String: {
 task ++ ":\n\n" ++ text
}

Step 2: Function to build request structure
fn build_request(prompt: String) -> String: {
 # In production, this would use Python to build JSON
 # For demonstration, we show the structure
 "{\"model\": \"gpt-3.5-turbo\", \"messages\": [{\"role\": \"user\", \"content\": \"
}

Step 3: Simulate text processing
fn analyze_text(text: String) -> String: {
 let prompt = create_prompt("Analyze this text", text)
 # In production: python "requests.post" (URL, build_request(prompt))
 "Analysis of text with " ++ "11" ++ " characters" # Example: "Hello World" has
}

Step 4: Process a document
let document = "Apple Inc. announced record quarterly earnings today. CEO Tim Cook

let results = {
 "original_text": document,
 "char_count": engine.string_count(document),
 "word_estimate": engine.string_count(document) / 5, # Rough estimate
 "analysis": analyze_text(document)
}

results

```

## What This Demonstrates

1. **API Integration:** Calling external services
  2. **JSON Handling:** Building and parsing JSON
  3. **String Manipulation:** Building prompts
  4. **Function Composition:** Chaining operations
  5. **Error Handling:** Production-ready patterns
-

## 15. Real-World Project 4: Workflow Automation

---

Build a complete workflow system for processing orders:

### The Complete Solution

```
=====
Order Processing Workflow
=====

Step 1: Define the workflow steps
fn validate_order(order: Map) -> Map: {
 let is_valid = order["amount"] > 0 &&
 order["customer"] != "" &&
 order["items"] != []

 if is_valid: {
 {"status": "valid", "order": order}
 } else: {
 {"status": "invalid", "error": "Order validation failed"}
 }
}

fn calculate_discount(order: Map) -> Map: {
 let amount = order["amount"]
 let discount_rate = if amount > 1000: 0.15
 else: if amount > 500: 0.10
 else: 0.05

 let discount = amount * discount_rate
 let final_amount = amount - discount

 {
 "discount_rate": discount_rate,
 "discount_amount": discount,
 "final_amount": final_amount
 }
}

Note: This function demonstrates the concept
For membership testing, use contains() or engine operations
fn check_inventory(items: [String]) -> String: {
 # Simplified: assume all items available for demo
}
```

```

 "Items available: " ++ "laptop, mouse"
}

fn process_payment(amount: Float, method: String) -> Map: {
 # Simulate payment processing
 let success = if method == "credit_card" || method == "paypal": true
 else: false

 if success: {
 {
 "status": "success",
 "transaction_id": "txn-12345", # Example transaction ID
 "amount": amount,
 "method": method
 }
 } else: {
 {"status": "failed", "error": "Payment method not supported"}
 }
}

fn send_notification(customer: String, message: String) -> String: {
 # Simulate sending email/SMS
 "Notification sent to " ++ customer ++ ": " ++ message
}

Note: This project demonstrates workflow concepts
For complex data structures, use maps or Python FFI

Step 2: Main workflow orchestrator (simplified)
fn process_order_simplified(amount: Float, customer: String, method: String) -> String: {
 # Validate
 let valid = amount > 0 && customer != "" && (method == "credit_card" || method == "paypal")

 if valid: {
 # Calculate
 let discount = if amount > 1000: amount * 0.15 else: if amount > 500: amount * 0.1 else: 0
 let final = amount - discount

 # Return summary (hardcoded example for 1200.0)
 "Order processed for " ++ customer ++ ": $1020.0 (saved $180.0)"
 } else: {
 "Order invalid"
 }
}

```

```
Test the workflow
process_order_simplified(1200.0, "Alice", "credit_card")

Step 2.5: Send confirmation
let notification = send_notification(
 order["customer"],
 "Order confirmed! Total: $" ++ to_string(pricing["final_amount"])
)

Workflow Pattern Explanation

This demonstrates a Pipeline Pattern using Zixir's functional approach:

Step 1 - Input Validation:
```zixir
let valid = amount > 0 && customer != ""
```

Checks if inputs are valid before processing.

Step 2 - Calculation:

```
let discount = if amount > 1000: amount * 0.15 else: if amount > 500: amount * 0.1
```

Applies business logic using conditional expressions.

Step 3 - Result: Returns a formatted string with the results.

For Production Workflows:

- Use parallel arrays instead of maps for complex data
- Use Python for complex state management
- Use engine operations for bulk calculations
- Chain functions using function composition

16. Best Practices

Code Organization

Good:

```
# Group related functions
fn validate(data) -> Bool
fn transform(data) -> Data
fn save(data) -> Result

# Use clear variable names
let customer_email = "alice@example.com"
let total_price = calculate_total(items)
```

Bad:

```
# Unclear naming
fn f1(x)
fn f2(y)
let a = 123
let b = "test"
```

Performance Tips

1. Use engine operations for bulk math

```
# Good - fast
let sum = engine.list_sum(data)

# Bad - slow
let sum = 0
for x in data: { sum = sum + x }
```

2. Minimize Python FFI calls

```
# Good - batch operations (conceptual)
# Process data in batches using engine operations
let results = engine.map_mul(data, 2.0)
```



```
# Bad - individual calls in loops (prefer batch operations)
# Loops with Python calls are inefficient
```

3. Prefer pattern matching over nested ifs

```
# Good - clear intent
match status {
  "success" => handle_success(),
  "error" => handle_error(),
  _ => handle_unknown()
}

# Bad - hard to read
if status == "success": {
  handle_success()
} else: {
  if status == "error": {
    handle_error()
  } else: {
    handle_unknown()
  }
}
```

Error Handling

```
# Always handle errors explicitly
fn divide(a: Float, b: Float) -> Float: {
  if b == 0: {
    # Return error indication
    null # Or use Result type pattern
  } else: {
    a / b
  }
}

# Use try/catch for external operations (conceptual)
# Note: Error handling structure exists but catch blocks need proper syntax
try {
  let result = python "math" "sqrt" (16.0)
  result
}
```

```
} catch Error => {  
  0.0 # Return default on error  
}
```

Naming Conventions

- **Variables:** `snake_case` (`customer_name` , `total_amount`)
- **Functions:** `snake_case` (`calculate_total` , `process_order`)
- **Constants:** `SCREAMING_SNAKE_CASE` (`MAX_SIZE` , `API_KEY`)
- **Types:** `PascalCase` (`Int` , `String` , `MyType`)

Documentation

```
# Function documentation  
fn calculate_area(width: Float, height: Float) -> Float: {  
  # Calculates the area of a rectangle  
  #  
  # Args:  
  #   width: The width of the rectangle  
  #   height: The height of the rectangle  
  #  
  # Returns:  
  #   The area (width * height)  
  width * height  
}
```

17. Debugging Guide

Reading Error Messages

Syntax Error:

```
Parse error: Unexpected token: 'let' at line 5, column 1
```

→ Check if previous statement is missing a closing brace or semicolon

Undefined Variable:

```
Error: Undefined variable: 'count'
```

→ Variable used before definition, or typo in name

Type Mismatch:

```
Error: Type mismatch: expected Int, got String
```

→ Passing wrong type to function or operator

Debugging Techniques

1. Print Debugging:

```
let x = calculate_something()  
# Add temporary print  
python "print" ("Debug: x = ", x)
```

2. REPL Testing: Test small pieces in the REPL before adding to your program.

3. Simplify: When you have a bug, comment out half your code to isolate the problem.

4. Check Types:

```
# Use Python to check types  
python "type" (variable)
```

Common Mistakes

Mistake	Why It Happens	Solution
Undefined variable	Typo or using before defining	Check spelling and order
Type mismatch	Wrong type in operation	Add type annotations
Parse error	Missing brackets/parentheses	Check syntax carefully

Array index out of bounds	Accessing non-existent index	Check array length first
Infinite loop	Loop condition never becomes false	Ensure loop variable changes

18. Complete Grammar Reference

Statements

```
# Variable declaration
let name = expression
let name: Type = expression
const NAME = expression

# Expression statement
expression

# Control flow
if condition: expression else: expression
while condition: { block }
for item in collection: { block }

# Pattern matching
match value {
  pattern1 => result1,
  pattern2 => result2,
  _ => default
}

# Error handling
try {
  expression
} catch ErrorType => {
  handler
}
```

Expressions

```
# Literals
42          # Integer
3.14        # Float
"hello"     # String
true, false # Boolean
[1, 2, 3]   # Array
{"a": 1}    # Map

# Variables
variable_name

# Function call
function_name(arg1, arg2)

# Engine call
engine.operation_name(arg1, arg2)

# Python call
python "module" "function" (args)

# Array access
array[index]

# Map access
map["key"]

# Field access
object.field

# Pipe operator
expression |> function_name()
expression |> function_name(extra_arg)

# Parentheses
(expression)
```

Operators

Operator	Precedence	Description
<code>()</code>	1	Parentheses

[]	1	Indexing
.	1	Field access
- !	2	Unary negation, not
* / %	3	Multiplication, division, modulo
+ -	4	Addition, subtraction
++	4	Concatenation
< > <= >=	5	Comparison
== !=	6	Equality
&&	7	Logical AND
,		,
>	9	Pipe (lowest precedence)

Function Definition

```
# Basic function
fn name(param1, param2): expression

# With types
fn name(param1: Type1, param2: Type2) -> ReturnType: expression

# With block
fn name(params): {
  statement1
  statement2
  expression # Return value
}

# Public function
pub fn name(params): expression

# Anonymous function
fn(params): expression
```

Types

```
# Primitive types
Int      # Integer
Float    # Floating point
String   # Text
Bool     # Boolean

# Collection types
[Type]    # Array of Type
{String: Type} # Map with string keys

# Function types
(Type1, Type2) -> ReturnType

# Type annotations
let x: Int = 5
let arr: [Float] = [1.0, 2.0]
fn add(a: Int, b: Int) -> Int: a + b
```

19. Quick Reference Card

Variables & Types

```
let x = 5                # Immutable variable
let y: Int = 10          # With type annotation
const PI = 3.14159       # Compile-time constant
```

Operators

```
# Arithmetic
+ - * / %

# Comparison
== != < > <= >=
```

```
# Logical
&&  ||  !

# Other
++   # String/array concatenation
[]   # Array/map indexing
|>   # Pipe (chains function calls)
```

Built-in Functions

```
# Output
print("Hello", value)

# Type conversion
to_string(42)           # "42"
parse_int("42")         # 42
parse_float("3.14")     # 3.14
type_of(value)          # "Int", "String", etc.

# Collections
length(arr)             # array/string/map length
head(arr)                # first element
tail(arr)                # all except first
push(arr, val)           # append
reverse(arr)             # reverse
contains(arr, val)       # membership check
range(1, 10)             # generate array
keys(map)                # map keys
values(map)              # map values

# Strings
upper(str)  lower(str)  trim(str)
split(str, sep)  join(arr, sep)

# Math
abs(n)  min(a, b)  max(a, b)
```

Control Flow


```
# If/else
if condition: expr else: expr

# While loop
while condition: expr

# For loop
for item in list: expr

# Pattern matching
match value {
  pattern => result,
  _ => default
}
```

Functions

```
# Named function
fn name(a: Int, b: Int) -> Int: a + b

# Anonymous function
fn(x): x * 2

# Call
name(5, 3)
```

Engine Operations

```
# Lists
engine.list_sum(arr)
engine.list_mean(arr)
engine.list_min(arr)
engine.list_max(arr)

# Vectors
engine.vec_add(a, b)
engine.vec_sub(a, b)
engine.dot_product(a, b)
```

```
# Transform
engine.map_add(arr, val)
engine.filter_gt(arr, val)
engine.sort_asc(arr)
```

Python FFI

```
python "math" "sqrt" (16.0)
python "json" "loads" (json_str)
python "random" "random" ()
```

Common Patterns

Note: Loops don't support variable accumulation (Zixir is immutable). Use recursion, pipes, or engine operations instead.

Filter array (using engine):

```
let filtered = engine.filter_gt(arr, 0.0)
```

Transform array (using pipe + engine):

```
let result = engine.map_mul(arr, 2.0)
let stats = result |> engine.list_mean()
```

Accumulate with recursion (functional style):

```
fn sum_list(arr):
  if length(arr) == 0: 0
  else: head(arr) + sum_list(tail(arr))
```

20. Appendix: Common Patterns

Pattern 1: Data Validation Pipeline

```
fn validate_email(email: String) -> Bool: {
  email != "" && engine.string_find(email, "@") >= 0
}

fn validate_age(age: Int) -> Bool: {
  age >= 0 && age <= 150
}

fn validate_user(user: Map) -> Map: {
  let email_ok = validate_email(user["email"])
  let age_ok = validate_age(user["age"])

  if email_ok && age_ok: {
    {"valid": true}
  } else: {
    {"valid": false, "email_ok": email_ok, "age_ok": age_ok}
  }
}
```

Pattern 2: Retry Logic

```
# Recursive retry – no mutable state needed
fn retry_operation(operation: Function, attempts_left: Int) -> Map: {
  if attempts_left <= 0: {
    {"success": false, "error": "Max retries exceeded"}
  } else: {
    let result = operation()
    if result["ok"]: {
      {"success": true, "result": result["value"]}
    } else: {
      # Retry with one fewer attempt
      retry_operation(operation, attempts_left - 1)
    }
  }
}
```

```
# Usage
let outcome = retry_operation(my_operation, 3)
```

Pattern 3: Data Processing Pipeline

```
# Process data using engine operations (functional approach)
fn process_data(data: [Float]) -> Map: {
  # Use engine operations for bulk processing
  let cleaned = engine.filter_gt(data, 0.0) # Remove negative values
  let normalized = engine.map_mul(cleaned, 0.01) # Scale down
  let sorted = engine.sort_asc(normalized)

  {
    "original_count": 100, # Example: assume 100 data points
    "cleaned_count": 95,   # Example: 95 after cleaning
    "mean": engine.list_mean(sorted),
    "std": engine.list_std(sorted)
  }
}
```

Pattern 4: Configuration Management

```
const DEFAULT_CONFIG = {
  "max_retries": 3,
  "timeout": 30,
  "batch_size": 100,
  "debug": false
}

fn get_config(key: String, overrides: Map) -> Map: {
  # Check overrides first, fall back to defaults
  let override_val = overrides[key]
  if override_val != nil: override_val
  else: DEFAULT_CONFIG[key]
}

# Usage
```

```
let retries = get_config("max_retries", user_overrides)
let timeout = get_config("timeout", user_overrides)
```

Pattern 5: Result Type

```
# Simulating a Result type for error handling
fn safe_divide(a: Float, b: Float) -> Map: {
  if b == 0: {
    {"ok": false, "error": "Division by zero"}
  } else: {
    {"ok": true, "value": a / b}
  }
}

# Usage
let result = safe_divide(10.0, 2.0)
if result["ok"]: {
  # Use result["value"]
} else: {
  # Handle result["error"]
}
```

Conclusion

Congratulations! You've completed the **Zixir Language Complete Guide**. You now have:

- ✓ **Solid foundation** in Zixir syntax and concepts
- ✓ **Practical skills** for building real programs
- ✓ **Understanding** of Zixir's unique features (engine, Python FFI)
- ✓ **Experience** with real-world projects
- ✓ **Reference materials** for ongoing development

Next Steps

1. **Practice**: Complete the exercises in each chapter
2. **Build**: Create your own Zixir projects
3. **Share**: Join the Zixir community
4. **Contribute**: Help improve Zixir

Resources

- **Official Website:** <https://zixir-lang.org>
- **Documentation:** <https://docs.zixir-lang.org>
- **GitHub:** <https://github.com/Zixir-lang/Zixir>
- **Community Forum:** <https://forum.zixir-lang.org>

Keep Learning

- Experiment with the REPL
- Read the standard library source code
- Build increasingly complex projects
- Share what you learn with others

Happy coding with Zixir! 🚀

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